

Home Lighting's Tech Makeover

Beyond large-scale and specialised smart lighting implementations for achieving the right ambience and energy savings lies a market that captures the hearts of home owners



Philips Hue

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Smart lighting technology is on a high! From Internet-connected lights and 3D-printed light emitting diode (LED) optics to light-emitting electrochemical cells and hybrid organic-inorganic LEDs, there is no end to tech innovations in this space. Mind-boggling applications are also being demonstrated and deployed around the world, such as age-specific lighting at retirement homes, supportive lighting at healthcare centres and smart lighting to improve acoustic comfort at opera houses, to enhance the appeal of exhibits at art workshops and deepen devotion at chapels.

Across the world, local municipalities are working out how smart

lighting can help save energy for their communities, with some having even adopted such solutions. In India too, Philips Lighting, together with the

Climate Group, demonstrated to municipal corporations in Kolkata how they could reduce urban emissions and save energy from street lighting. The team showed that LED lighting consumes about 40 to 50 per cent less energy as compared to conventional sodium vapour lamps, and if fitted with smart adaptive controls, the savings could be even higher, in the range of 70 to 80 per cent. Around 300 street lights in the city were then retro-fitted with Philips Roadstar LED streetlights.

While the use of smart lighting products in specialised environments and other public spaces is justified by the savings and other benefits, for a home user it still appears to be a costly proposition. The availability of simple, easy-to-install individual smart light bulbs for home users is also quite low as compared to larger solutions. When users compare the \$50+ cost of installing a smart light bulb with the cost of installing an ordinary light bulb, the choice is clearly in favour of the latter, unless they have a very pressing need for smart lighting. Players

Smart bulbs: A quick primer

- Smart LED bulbs will fit into regular bulb sockets.
- Rough specifications of the LED bulbs: 800+ lumens, 9-12 watts energy consumption, 65-85 lumens/watt efficiency and a life of around 25,000 hours.
- Smart bulbs include processor, memory, real-time clock, communication technology and, sometimes, a heat sink.
- Most Zigbee based bulbs require a hub to connect with your home Wi-Fi network. The hub will come with the bulbs, as a kit, and will need to be plugged into a power socket. No other complex installation would be required.
- Bulbs built-in with Bluetooth or Wi-Fi do not require a hub.
- You need to download the associated app on your smartphone or tablet. It will automatically detect your hub and bulbs.
- You can then control, schedule and program your bulbs. It is as easy as fit bulb, plug-in hub, download app, and control is in your hands!
- Most apps can control around 50 bulbs, so you can add more bulbs on-the-go.
- Prices vary from \$50 to \$200 per kit, which includes two to three bulbs, and a hub, where required. Individual bulbs cost between \$15 and \$100.